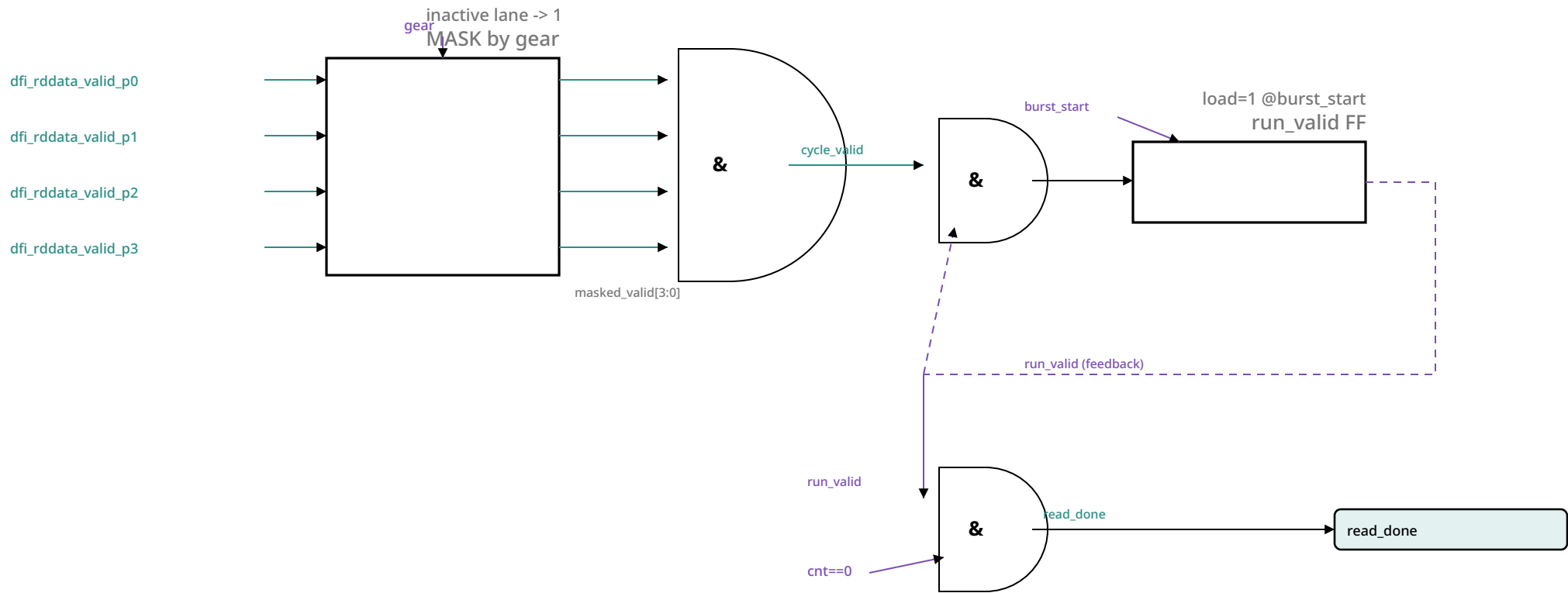


Read valid contiguity (no-bubble guarantee)



`cycle_valid` = AND of ACTIVE-phase valids (gear masks inactive lanes to 1): `1:1=vp0` ; `1:2=vp0&vp1` ; `1:4=vp0&..&vp3`.
`run_valid` <= `run_valid` & `cycle_valid` ; reset to 1 at `burst_start`. A bubble (any active-cycle valid low) drops `run_valid`.
`read_done` = `run_valid` & (`cnt==0`). Bubble -> `run_valid=0` -> `read_done` blocked (flag as read error). Contiguous -> done.